

Harsh Thomare

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Education

University of California, Irvine Dec 2024
B.S., Computer Science and Engineering, *cum laude*, GPA 3.80/4.00

Publications

- Yonatan Gizachew Achamyelch, **Harsh Thomare**, Mohammad Abdullah Al Faruque. “HELIOS: Hierarchical Graph Abstraction for Structure-Aware LLM Decompilation.” *arXiv:2601.14598*, Jan 2026. Presented at the LAST-X Workshop, NDSS Symposium 2026.

Research Experience

Research Assistant May 2025 – Nov 2025
UC Irvine, AI Cyber-Physical Systems (AICPS) Lab

- Co-developed HELIOS, a structure-aware LLM decompilation method that gives the model a hierarchical text view of Ghidra-extracted control-flow graphs instead of flat decompiler output
- Validated HELIOS across six ISAs at -O0 through -O3; on x86_64, compiler feedback reached 94.9–96.5% compilability and 53.2–55.9% functional correctness, beating fine-tuned LLM4Decompile-6.7B (36.3%) with no training
- Did the literature review and co-wrote paper sections

Research Collaborator Mar 2026 – Present
UCLA Security Lab

- Building an agentic benchmark testing whether LLM agents can recover hidden behavior of transient-execution programs from hardware measurements (unpublished, in progress)

Undergraduate Researcher Sep 2022 – Mar 2023
UC Irvine, AI Cyber-Physical Systems (AICPS) Lab

- Built Raspberry Pi automation for recording differential pressure-sensor data in an acoustic side-channel attack project
- Reviewed prior work on acoustic side channels and ran early evaluations of candidate speech-recovery models on the recorded sensor data
- Early-stage contributions acknowledged by name in “A Fly on the Wall: Exploiting Acoustic Side-Channels in Differential Pressure Sensors,” ACSAC 2024

Work Experience

Embedded Software Engineer Jul 2024 – Present
Mainspring Energy, Menlo Park, CA

- Built Docker-based OTA release pipeline publishing signed firmware images to a CDN, serving delta updates with resume support across 1,000+ field devices
- Wrote a C UART bootloader with A/B image banks, CRC32 integrity checks, and automatic rollback on failed boot
- Implemented sensorless back-EMF control for 60kW motors on bare-metal STM32F4, raising stopping force by 41% and cutting thermal buildup by 30%

Electrical Engineering Intern

Jan 2024 – Apr 2024

Second Order Effects

- Designed Verilog data-capture engine on FPGA for high-speed ADC signal acquisition with full-coverage SystemVerilog testbenches
- Wrote Linux device driver exposing captured samples via sysfs for system-level verification

Projects

Qwen Mega-Kernel: Fused CUDA/Triton Kernels

- Fused Triton/CUDA kernels for Qwen transformer blocks and profiled them on an NVIDIA A6000 with Nsight Compute, lifting achieved memory bandwidth by 30%

IEEE Micromouse

2021 – 2022

- Designed the PCB for the maze-solving micromouse robot
- Built BLE-based wireless logging to stream telemetry for PID tuning

Teaching

Workshop Lead, IEEE Keebs Club

- Founded the workshop program: came up with the idea, designed the curriculum, and handled parts procurement
- Taught soldering and PCB design to 60 students through hands-on mechanical-keyboard build workshops

Honors and Awards

- *Cum laude*, B.S. Computer Science and Engineering, UC Irvine, 2024
- UROP Distinguished Research Fellowship, UC Irvine, 2023
- Sigma Prize Winner, Mental Health Innovation Challenge

Service

Ambassador

2016 – Present

Akshaya Patra Foundation

- Led a Bay Area sapling-sale initiative, coordinating distribution across volunteers; raised funds to feed 5,000 children

Relevant Coursework

- COMPSCI 190 – Efficient ML Computing (A)
- COMPSCI 143A – Principles of Operating Systems (A)
- COMPSCI 142A – Compilers and Interpreters (A)
- EECS 112 – Organization of Digital Computers (A)